Literature Review

Paper Details- Title

Study of the Hydrous Ethanol Purification Stage: Simulation and Sensitivity Analysis of Extractive Distillation Using DWSIM Simulator

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Background

Hydrous ethanol produced during fermentation contains a significant amount of water. Since ethanol and water form an azeotropic mixture, conventional distillation cannot produce fuel-grade ethanol. Industrial production therefore employs **extractive distillation**, where a high-boiling solvent such as **ethylene glycol (EG)** selectively alters the vapor-liquid equilibrium and enables ethanol purification.

Objective of the Paper

The objective of the study was to:

- simulate an industrial ethanol purification process using DWSIM,
- perform sensitivity analysis on major operating variables,
- determine operating conditions that maximize ethanol purity,
- minimize energy consumption,
- improve extractive distillation performance.

Process Description

The process consists of two distillation columns:

Extraction Column

Purpose:

- Separate ethanol from the ethanol-water azeotrope

- Ethylene glycol acts as entrainer
- Ethanol leaves from the top
- Water and EG leave from the bottom

Regeneration Column

Purpose:

- Recover ethylene glycol
- Remove remaining water
- Recycle EG back to Column 1

Sensitivity Analysis Performed

The authors investigated the influence of:

- Solvent mass flow rate
- Solvent temperature
- Solvent pressure
- Feed pressure
- Feed temperature
- Feed flow rate
- Number of stages

Each parameter was varied individually while observing:

- ethanol purity
- reboiler duty
- condenser duty
- process energy consumption
- product composition.

Major Findings

The paper concluded that:

- Higher solvent temperatures reduced reboiler energy demand.
- Approximately **197°C solvent temperature** produced favourable performance.
- Around **4000 kg/h solvent flow** improved ethanol purification.
- Atmospheric solvent pressure was sufficient.

- Twenty theoretical stages provided better separation.
- Regeneration column achieved approximately:
    - **99% ethylene glycol recovery**
    - **98.5% water recovery**
- Extraction column achieved approximately:
    - **98.86 wt% ethanol purity**
- Reboiler duty reduced to approximately **−2237 kW** under optimum conditions.

Limitations Identified

While reproducing this work, several practical limitations were identified.

The paper reports sensitivity plots but does not provide a complete operating dataset for every simulation. Several design specifications required to reproduce the model are either omitted or insufficiently described. As a result, the published work cannot be directly replicated without making additional engineering assumptions.

The study also optimizes one variable at a time, whereas real industrial operation requires simultaneous adjustment of multiple interacting variables.

Motivation for the Present Work

This project extends the published work by:

- rebuilding the extractive distillation process from scratch in ChemSep,
- identifying missing process specifications,
- determining feasible operating ranges through convergence studies,
- generating a structured simulation dataset,
- developing machine learning models capable of predicting process performance,
- creating an AI-assisted optimization framework for rapid operating-condition selection.

Rather than replacing rigorous process simulation, the developed digital twin serves as a decision-support tool that significantly reduces the number of simulations required during process optimization.